AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

PERSONAL DETAILS:

Name:	Afiso Nyambi
Staff No.:	63
Job Title:	Lab Manager

ASSESSMENT:

Attempt:	1st
Date:	08 July
Time:	12H00

ASSESSORS:

Name & Staff nr.:	1.	
	2.	
	3.	
	4.	
Observers: (optional)	1.	2.
Translator: (if required)		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **Determination of Volatile Matter in Coal** **Desired Outcome:** At the end of this module the employee should be able to perform volatile matter determination in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	The competency and development tool is based on ISO 562
3.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Volatile Matter determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
4.	You are going to demonstrate the use of determination of the Volatile Matter. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **27** points of which **24** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence. There is a set time limit of **60** minutes for the completion of the assessment
5.	You have the right to an interpreter and the right to have an observer present during the assessment.
6.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

Assessment for Volatile Matter

Date: 08/09/25

Candidate name: Sipho **Staff No:** 03

Candidate Signature: [signature]

No	Question:	Points	Score
1.	At what temperature is the test sample incinerated? 900±5°C ✓	* 1	1
2.	What does one use to monitor the time in the furnace? Stopwatch ✓	* 1	1
3.	What is the maximum time for incineration of the test sample? 7±2secs. ✓	* 1	1
4.	To what temperature must the volatile furnace recover after 4 minutes? 896°C ✓	* 1	1
5.	Where does one cool the crucible and stands? bench & dessicator ✓	1	1
6.	Complete the sentence: If the furnace has not recovered back to 896°C within the 4th minute, discard the samples and do repeats. ✓	* 2	2
7.	What is meant by repeatability? Analysis conducted by same analyst using same equipment in the same day and place over a period of time. ✓	* 3	3
8.	How often drifts are checks run? After 4 samples. ✓	* 1	✓

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9.	How often and by whom are the verification checks done? After 4 runs by lab technician.	* 2	2
10.	Between which masses are the masses for volatile determination controlled? 0.95~1.05g.	* 1	1
11.	The insertion time of a volatile stand can be 10 seconds. True or False? False.	* 1	1
12.	A volatile crucible stand can be kept in the furnace for 7min± 8 seconds. True or False? False; 7min ± 8secs.	* 1	1
13.	How does one identify volatile crucible and lids? Number has to be the same across both	* 2	2
14.	What is the repeatability limit for volatile determination? 0.3% VM < 10% and 3% of ave VM > 10%	* 1	1
15.	What is subtracted after testing to get volatile results? Inherent moisture.	* 1	1
16.	How many samples must be used in the repeat analyses? Duplicate	1	1
17.	How often are crucibles cleaned? Every after use.	* 1	1
18.	What should the distance between the crucible and the lid? 38mm height x 25mm OD.	1	1
19.	How does one ensure that new lids fit their crucible? check for 0.5mm.	* 1	
20.	What laboratory documents are to be used when	*	

	determining the Volatile Matter AS013TF. ✓	1	1
21	Are you allowed to carry out volatile analysis with vacant places in your stand?	* 2	
	No. all spaces should be filled. ✓		
	Total:	27	27

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorisation.**

Assessor Name N. Mabi **Signature:** _____ **Date:** 8/9/25

Learner Signature: Fiso Myambi **Date:** 08/09/2025

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
		1611	25.8+3.70 = 29.5	29.35	29.34
Acceptance Tolerance: 0.3% or 4% of average between duplicate results					

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Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

PERSONAL DETAILS:

Name:	Sifiso Myambi
Job Title:	Laboratory Manager

ASSESSMENT:

Attempt:	1st
Date:	08 Sept 2025
Time:	13H00

ASSESSORS:

Name	1.	
	2.	

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Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **the Determination of Inherent Moisture** **Desired Outcome:** At the end of this module the employee should be able to perform Determination of inherent Moisture in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with Inherent Moisture determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the inherent moisture During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **26** points of which **16** are critical (*). To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

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Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Assessment Inherent Moisture

Date: 03/11

Candidate name: Brijesh **Staff No:** 03

Candidate Signature: [signature]

No	Question:	Points	Score
1.	At what temperature is the test sample analysed?	* 1	1
	107°C ✓		
2.	Who is responsible for plotting control charts?	* 1	1
	lab technician. ✓		
3.	Who is responsible for investigating any deviation?	* 1	1
	lab tech/supervisor/manager ✓		
4.	What is meant by the term repeatability?	2	2
5.	What does high inherent moisture content indicate?	1	1
	Poor rank. ✓		
6.	Complete the sentence: Oxidized Coals become porous which increases the surface area inside allowing more oxygen in increasing the combustion rate. ✓	* 2	2
7.	Where must the crucibles be cooled and for how long?	* 2	2
	Benchtop and desicator, 10 min each. ✓		
8.	What accompanies each batch of samples for inherent moisture?	* 1	
	Control sample. ✓		1
9.	How must the crucible lid be? Uncracked and clean. ✓	1	1

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10.	Name the one desiccant that can be used? Silica gel. ✓	1	1
11.	The inherent moisture sample is incinerated for one hour. True or False? False. ✓	* 1	1
12.	The inherent moisture crucible is weighed back while still hot. True or False? False. ✓	* 1	1
13.	Inherent moisture crucibles are cooled with the lid open True or False? False. ✓	* 1	1
14.	What is the function of silica gel in the dessicators? To absorb heat/moisture. ✓	1	1
15.	How does one regenerate silica gel? Insert into the drying oven. ✓	1	1
16.	How many samples must be used in the repeat analyses? Duplicate. ✓	1	1
17.	Within which limits must mass be controlled? 0.95 - 1.05g ✓	* 1	1
18.	What are the repeatability limits for inherent moisture? 0.1% of IM < 5% ✓ ✓ and 0.1% for IM > 5% ✓	* 4	4
19.	Inherent moisture can only be done when the balance has been verified True/ False False ✓	* 1	1
20.	What is the relation between inherent moisture and volatile matter? Corr VM = VM - IM ✓	1	1
	Total:	26	25

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Comments by Assessor(s): Candidate is deemed competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorization.**

Assessor Name N. Mohd **Signature:** _____ **Date:** 8/9/25

Learner Signature: _____ Date: 08/09/2025

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
		1611	~~18.5~~	~~17.84~~	~~18.05~~
			3.70	3.55	3.51

Acceptance Tolerance: 0.15% between duplicate results for IM >5% and 0.1 for IM ≤5%

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Subject	TECHNICAL SIGNATORY		

TECHNICAL SIGNATORY TRAINING AND ASSESSMENT

Name and Surname: Afiso Nyambi

Date: 13/11/2025

Total = 50 marks Questions = 30

SECTION A: Roles, Responsibilities and SANAS Requirements (R03-10 & R04) (12 marks)

1. Explain the primary responsibilities of a Technical Signatory as outlined in R03-10. How does this role ensure the technical validity of test results? (2)

Answer: Technical signatories authorise the release of test reports or calibration results. They confirm that all methods, equipment and personnel comply with ISO 17025 requirements. (2)

2. Describe the process your laboratory follows to nominate and approve a new Technical Signatory. Which documentation and evidence must be submitted to SANAS? (2)

Answer: The lab nominates a new TS through internal approval processes that includes competency evaluations, witnessing and management approval. Documents submitted to SANAS include CV, qualifications, witness records and practical evaluation. (2)

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3. Discuss three qualification or experience requirements that must be met before approval as a Technical Signatory under R03-10. (2)

Answer: (2)
⇒ Relevant qualification ✓
⇒ Sound knowledge and understanding of ISO Standards and test methods ✓

4. Explain the difference between a Nominated Representative and a Technical Signatory according to R03-10, and why these roles must be clearly defined. (2)

Answer: (2)
NR is person who allows for communication btw SANAS and the CAB while a TS ensures that all results given to clients have gone through all processes that ensure conformity.

5. Describe how the Technical Signatory ensures correct use of the SANAS logo and reference to accreditation in reports, per R04. Give one example of incorrect usage. (2)

Answer: (2)
SANAS logo and accreditation reference only applies to methods that have been assessed and accredited, these methods should be distinguishable from those not accredited and these should be marked and denoted on all test reports

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6. Outline the procedure for referencing SANAS accreditation on laboratory reports to maintain traceability. (2)

Answer:
* Test report should have accreditation number, method identifier which aligns with accreditation scope. ✓ (2)

SECTION B: Suspension, Withdrawal and Reinstatement (R51-10) (8 marks)

7. Under what circumstances may SANAS suspend or reduce a laboratory's scope? Provide an example. (2)

Answer:
If the CAB fails to meet SANAS requirements, e.g. loss of competency of critical staff. ✓ (2)

8. State the laboratory's obligations during suspension or withdrawal, covering both administrative and technical aspects. (2)

Answer:
The lab must stop using the SANAS logo, stop the release of accredited result and inform affected clients. ✓ (2)

9. Describe the process and evidence required for reinstatement following suspension. (2)

Answer:
Lab must submit corrective evidence addressing the root cause of non compliance. ✓ followed by SANAS reassessment. Only after the assessment can the lab be reaccredited (based on outcome of assessment) (2)

10. Discuss the possible impacts of accreditation withdrawal on laboratory credibility and client confidence. (2)

Answer:
- Undermines the credibility of the CAB
- Leads to the loss of clientele.
- Loss of funds.

SECTION C: Proficiency Testing and Interlaboratory Comparison (R80-06) (8 marks)

11. Explain the importance of proficiency testing (PT) participation for accredited laboratories and how it supports confidence in results. (2)

Answer:
A PT scheme allows for interlaboratory comparison, which means that a lab gets to see how it performs when compared to other labs. This is important because labs cannot be operated in isolation but rather the results and processes must be comparable (reproducibility) to those of other labs.

12. List criteria to consider when selecting a suitable PT or interlaboratory comparison provider. (2)

Answer:
- Provider must be competent/ISO 17043 accredited
- Use of similar methods and matrices.

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13. How should a Technical Signatory respond to unsatisfactory PT performance? Describe the steps for investigation and correction. (2)

(2)

Answer:
Open a non-conformance, investigate the root cause, Implement corrective actions, check methods and document the outcome of investigation and close the NC.

14. Discuss how PT data can be used for ongoing staff competence verification. (2)

(2)

Answer:
The results of PT form part of competency evaluations because they aid in confirming the staffs analytical proficiency. Successful performances imply that technical competencies are going well and there is less need to witness testing.

SECTION D: Calibration and Intermediate Checks (TR25) (10 marks)

15. Define "intermediate check" per TR25 and explain its purpose in maintaining measurement confidence. (2)

(2)

Answer:
Intermediate checks are performance checks carried out in b/w calibrations to verify ongoing accuracy and equipment stability. They ensure that whatever results the labs gets are still valid and accurate.

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16. Give three examples of coal-testing instruments requiring intermediate checks and describe what parameter each check verifies. (2)

Answer:
Drying oven: Probe / thermometer used to verify temp control

17. Explain the difference between calibration and verification, giving one laboratory example. (2)

Answer:
Calibration is checking the instruments readings against a certified reference material, e.g., CV calibration using benzoic tablet. On the other hand verification confirms the performance of instrument without changing the calibration

18. Describe how the Technical Signatory ensures calibration certificates meet traceability requirements in TR25. (2)

Answer:
⇒ Check for metrological traceability on the certificate, environmental conditions, uncertainties & uncertainty statements
⇒ Valid calibration dates and equipment no's on certificates should match those in the report/register

19. Discuss the impact of using equipment that has missed scheduled checks on test validity and accreditation compliance. (2)

Answer:
- Leads to invalidity of results since there is no guarantee that the equipment still work optimally.
- Leads to NCs, results may be rejected during audits.

(2)

SECTION E: Method Validation and Verification (TR26) (8 marks)

20. Differentiate between validation and verification of analytical methods as per TR26 and indicate when each applies. (2)

Answer:
Method validation refers to the act of confirming if a new method is suitable for its intended use. Verification basically confirms if methods perform as expected under certain conditions as set by the lab.

(2)

21. Identify three performance characteristics evaluated during method validation under TR26. (2)

Answer:
- Accuracy & Precision
- Uncertainty
- Ruggedness/Robustness.

(2)

22. Explain how the Technical Signatory ensures validated methods remain fit for purpose, citing two review mechanisms. (2)

Answer:
⇒ Check for performance of the CAB in PT scheme ✓
⇒ Use of control charts, CRMS ✓
⇒ Check the trends of QC results ✓
⇒ Use of standard reference material ✓

23. Describe the link between uncertainty of measurement and method validation, and its importance in coal testing decisions. (2)

Answer: There is no perfect system in the world/universe, all methods come with a certain level of uncertainty/errors. Uncertainty of measurement gives the degree to which results are within acceptable limits, dictates comparability among labs. Forms a part in presenting evidence that MV did take place.

SECTION F: Technical Competence and Laboratory Operations (4 marks)

24. Explain how the Technical Signatory ensures accuracy and repeatability in: Ash Determination, Calorific Value, Inherent Moisture, ISO Volatile Matter, and Sample Preparation. (2)

Answer:
⇒ Use of calibrated equipments & verified methods. ✓
⇒ Do replicate testing, use of control samples and evaluating competency of staff. ✓

25. Reflect on how SANAS documents R03-10 to TR26 collectively promote data integrity and continual improvement in an accredited coal laboratory. (2)

Answer: R03 is concerned with requirements for TS and NR, R04 outlines

the correct use of SANAS logos and symbols and what must be in reports. TR26 is concerned with MV. Together these ensure that equipments/methods/staff used to show the lab are competent and actually understand what needs to be done. Thus, promoting continuous improvement of QMS of the CAB.

SECTION G: Applied Analytical Calculations (Laboratory Data Exercise) (10 marks total – 5 questions × 2 marks)

26. Convert the Volatile Matter result from air-dry to dry-basis. (2)

Answer: VM = 27.62% (AD) ; IM = 3.83% ; Corrected VM = 27.62 − 3.83 = 23.79%

AD → DB: $DB = \frac{(100 \times AD)}{(100 - IM)} = \frac{(100 \times 23.79)}{(100 - 3.83)} = 24.74\%$

27. Calculate the Total Moisture %. (2)

Answer: TM = ?? ; SM = 4.26% ; RM = 3.01%

$RM = \frac{3.01(100 - 4.26)}{100} = 2.88\%$

$\Rightarrow TM = 4.26 + 2.88 = 7.14\%$

28. Convert the % Ash result from air-dry to as-received basis. (2)

Answer:

$AR(\%\ ASW) = \left(\frac{100 - TM}{100 - IM}\right) \times \%\ AD$

$= \frac{(100 - 7.14)}{(100 - 3.83)} \times 20.83 = 20.11\%$

29. Convert the Calorific Value from MJ/kg to kcal/kg. (2)

Answer: CV (MJ/kg) = 25.52 MJ/kg ✓ (2)

Kcal ⇒ 25.52 × 238.5 = 6099 kcal/kg ✓

30. Apply the appropriate correction for Volatile Matter and explain its purpose. (2)

Answer: VM is defined as everything removed from the coal besides the inherent moisture. Thus if one wants to find out how much volatile matter was driven off, they'll have to remove IM. (2)

Corrected VM = VM - IM ✓

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ASSESSOR'S MARKING SHEET

Use this section to record marks and comments for each question.

Question No.	Mark Awarded	Assessor Comments
1	2	
2	2	
3	2	
4	2	
5	2	
6	2	
7	2	
8	2	
9	2	
10	2	
11	2	
12	2	
13	2	
14	2	
15	2	
16	2	
17	2	
18	2	
19	2	
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21	2	
22	2	
23	2	
24	2	
25	2	
26	2	
27	2	
28	2	
29	2	
30	2	